

QUANTITATIVE CHEMISTRY – CONSERVATION OF MASS

KEY VOCABULARY

Conservation of mass: no atoms are made or lost in a reaction.

Balanced equation: same number of each atom on both sides.

Reactant: a substance that reacts (left of the arrow).

Product: a substance made (right of the arrow).

COMMON MISCONCEPTIONS

~~Mass is lost when a gas is given off.~~

✓ The gas escapes — but its mass is not destroyed.

~~You can change subscripts to balance.~~

✓ Only change the big numbers (coefficients).

~~Mass increases when wood burns.~~

✓ Mass seems to fall — gases escape into the air.

WORKED EXAMPLE

12 g of carbon reacts with 32 g of oxygen.

mass of products = mass of reactants

mass of CO₂ = 12 + 32

= 44 g

No atoms are lost — mass is conserved.

1 STATE THE LAW

State the law of conservation of mass. [1 mark]

2 CALCULATE A MASS

8 g of methane reacts with 32 g of oxygen to make 22 g of CO₂. Calculate the mass of water made. [2 marks]

Answer: _____ g

3 MASS DECREASE

Explain why the mass appears to decrease when a metal carbonate is heated. [2 marks]

4 MASS INCREASE

Explain why magnesium gains mass when burned in air. [2 marks]

5 BALANCE IT

Balance: ___ H₂ + O₂ → ___ H₂O. [2 marks]

Add the missing balancing numbers.

2

1

6 CLOSED SYSTEM

Explain why mass stays constant when a reaction is done in a sealed flask. [2 marks]
